

Additive Structure of Phonological Correspondences: A Manual

Thesis, methodology, operations, and findings — with worked examples in real words

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How to read this book. Most chapters follow the same teaching flow, signposted by bold headings: the motivating question, the intuition in plain language, the formal definition, a worked example in real words with IPA, the result in Indo-European and Austronesian, the limits (what *not* to conclude), and how to reproduce it. Start with Chapter 0, which walks the whole method by hand on real words. A standing epistemic rule, stated once and kept throughout. During the *discovery* phase this project uses no protoforms, no known sound laws, and no OAS classes as inputs. OAS is deliberately withheld: it would inject psychodynamic factors into a phase whose whole point is to find structure *before* interpretation. Like reconstructions and historical laws, OAS is reserved as an external partition for later contrast, never a coordinate of discovery. *Family-blind* comparison is likewise a future phase (aimed at the "universal" questions we are deliberately *not* asking yet), not a present result.

0.1 We start with words, not theories

Take one meaning — 'two' — and collect how a few Indo-European languages actually say it, each written in IPA (the phonetic alphabet). Nothing is reconstructed; these are attested, documented forms:

language	'two' (IPA)
Bulgarian	dva
Ancient Greek	duo
Serbo-Croatian	dva
Armenian (Eastern)	jerkʰu
Armenian (Western)	jergu
Yiddish	cvej

These words are cognates: descendants of the same ancient word, changed differently in each language. A historical linguist would now reconstruct the ancestor and derive each form from it. We will not. We will only compare the forms *to each other*.

0.2 Line two words up, letter by letter

Take the two Armenian varieties. Write their sounds in columns so that matching sounds sit together (this is called an alignment):

Eastern: j e r kʰ u
Western: j e r g u

Read down each column. Four columns are identical (j, e, r, u). Exactly one column differs: kʔ in the East, g in the West. That single difference is the raw material of everything that follows.

0.3 Say *what* differs, not *which becomes which*

kʔ and g are both sounds made at the back of the mouth (velars). The question a phonetician asks is: *in which properties — which “features” — do they differ?* Line up their features:

feature	kʔ	g	differ?
voiced (vocal cords vibrate)	no	yes	?
velar (back of mouth)	yes	yes	—
stop (full closure)	yes	yes	—
... (all others)	=	=	—

They differ in exactly one feature: voicing. So we write this correspondence as

$$‘kʔ’ \sim ‘g’ = \{voi\}.$$

That object — the *set of features that differ* — is what we call an operator. Notice what it is not: it does not say the sound “became voiced” or “became voiceless.” It is symmetric — it only says *these two sounds differ in voicing*. Direction (who came first) is a separate, later question. This single decision — describe a correspondence by *which features differ* — is the whole idea of the book.

0.4 A second example, with two features

Now ‘tooth’, and two more languages:

language	‘tooth’ (IPA)
Armenian (Eastern)	atʔam
Armenian (Western)	adam
Breton	dant
Modern Greek	ðodi

Align Eastern vs Western Armenian: a–tʔ–a–m against a–d–a–m. Again one column differs: tʔ d, and again the only feature is voicing → {voi}. The same operator, {voi}, shows up in a different word. That recurrence is the point: operators are not one-offs; they *repeat* across the vocabulary.

Now compare the Greek ð (the “th” in *this*) with a d: they differ in two features — a d is a full stop, a ð lets air flow (continuant) and has a certain hiss (strident). So

$$‘d’ \sim ‘ð’ = \{cont + strid\}.$$

An operator can bundle one, two, or three features. {voi} is an atom (one feature); {cont+strid} is a molecule (several).

0.5 Do this everywhere → the *repertoire*

Repeat §0.2–0.4 for every pair of cognate words in every meaning across the whole family. Collect all the operators you find, keeping the ones that recur often enough to be real (not alignment noise). The

resulting set is the family’s repertoire, written O . For Indo-European it has 67 distinct operators; for Austronesian, 22.

Here are Indo-European’s atoms — its one-feature operators — each shown as the *real sound pairs* that realize it in the data:

atom	what it is	real correspondences
{voi}	voicing	d t, g k, b p, s z, f v
{cont}	stop $\leftrightarrow$ fricative	t θ, d ð, k x, g ɣ, b β
{ant}	front $\leftrightarrow$ back sibilant	s ʃ, z ʒ
{strid}	plain $\leftrightarrow$ strident	s θ, z ð, v β
{hi}	palatalization / raising	l l ^j , n n ^j , k q
{back}	palatal $\leftrightarrow$ velar	c k, c ^h k

A reader who knows historical linguistics will recognize t θ, d ð, k x — this is the shape of what is traditionally called the Germanic consonant shift (Grimm’s Law). We did not put it there. It *fell out* of comparing attested forms. Whether it matches Grimm’s Law is a question we ask *afterward* (Part V), never an assumption we feed in.

0.6 Operators can be *added* — and the sum often lands back inside

Here is the first genuinely mathematical observation, and it is easy to see by hand. Suppose a family has the operator $b \ p = \{voi\}$ and the operator $p \ f = \{cont + strid\}$. What is the difference between b and f ? You *combine* the two: whatever changes once stays changed, whatever changes twice cancels. Here nothing cancels, so

$$\{voi\} + \{cont + strid\} = \{voi + cont + strid\},$$

which is exactly the operator $b \ f$. Combining operators is just XOR of feature-sets (symmetric difference). The striking empirical fact — the spine of this book — is that when you combine two operators from a family’s repertoire, the result is *another operator already in the repertoire* far more often than chance would predict. The repertoire is not a random grab-bag; it is additively organized. Measuring exactly how much, and against what “chance” means, is what Part III does.

0.7 What the rest of the book does

Everything from here is careful measurement of the object we just built by hand:

- How big is the space the repertoire lives in, and how much does it use? (Chapters 7, 9)
- Is the additive organization real, or an accident of the sounds involved? (Chapters 10, 17)
- Which operators are *missing* — combinations that could exist but don’t? (Chapter 12)
- Do related branches of a family share their repertoires more than random groups of languages? (Chapter 15)

If you followed §0.1–0.6, you already understand the method. The rest is rigor.

PART I — The idea and where it stands

1. The problem: patterns traditional linguistics has not given us

The question. Historical linguistics is one of the most successful sciences of the humanities. It reconstructs ancestral languages, states sound laws, and draws family trees, all from patient comparison of documented forms. So what is *missing*? What could an explicitly *mathematical* description of sound correspondences add that the comparative method has not already given us?

The intuition. The comparative method answers questions of the form “*how did these forms descend from their common ancestor?*” It is a method of inference toward a hidden past. But there is a different question it was never designed to ask: “*treated purely as a set of differences, what shape does the space of attested correspondences have?*” Is that space full or sparse? Does it cluster? Are some differences combinations of others? Do two families occupy the same region of the space, or different regions? These are questions about the geometry and algebra of the differences themselves, independent of which form came first.

The reframing. Write each sound correspondence not as “*segment a becomes segment b*” but as the set of phonological features in which a and b differ. That object — call it $\Delta(a, b)$ — is a vector. A whole family, then, contributes a *set* of such vectors: its repertoire O . Once we have a set of vectors we can ask genuinely new questions: its dimension (rank), how much of the space it can generate it actually uses (occupancy), whether combining two of its elements tends to land back inside it (additive structure), and which generable differences it pointedly avoids (the *holes* $\langle O \rangle \setminus O$). None of these are questions the reconstructive order poses, because that order is busy with a different task.

The limit, stated up front. This is not a replacement for the comparative method, and it is not a claim that “sound change is XOR.” It is a change of *order of inference* (Chapter 3). The mathematics describes attested differences; whether any of it lines up with reconstructions, laws, or family trees is a *separate* question, asked afterward, with objects that never entered the computation.

What you will be able to do. By the end of this book you will be able to take any language family with IPA-segmented word lists, compute its correspondence repertoire without reconstructing anything, measure its geometry and additive structure against explicit null models, and compare families — all reproducibly, one make target at a time.

2. The thesis in plain language

The one-sentence thesis. *From documented forms, phonological correspondences produce sparse repertoires whose additive, geometric, and distributional organization can be measured without taking an ancestral reconstruction as the origin of coordinates, and only afterwards contrasted with the regularities historical linguistics proposes.*

Unpacking it.

- *From documented forms* — the inputs are attested words with IPA segments, nothing reconstructed.
- *Sparse repertoires* — of everything the feature representation could express, a family uses very little... but, as we will see (Chapter 9), that sparsity is mostly an artifact of how vast the algebraic space is; measured against what the corpus actually makes available, the repertoire is fairly *dense*.

- *Additive organization* — combining two operators (XOR of their feature-differences) lands back inside the repertoire far more often than chance (Chapters 6, 10). The set behaves like an approximate subspace.
- *Measured without protoforms* — the computation uses only attested-to-attested alignments.
- *Contrasted afterwards* — reconstructions, laws, and family trees are brought in only to interpret and test what was found, never to build it.

A first taste in real words. Between Eastern and Western Armenian for 'two' (*erku*), $k^?$ corresponds to g : the single feature that differs is voicing, so the operator is $\{voi\}$. Between a Greek d and a Slavic z for 'tooth', two features differ — continuancy and stridency — so the operator is $\{cont+strid\}$. These are not directions of change; they are symmetric signatures of difference. The whole method is built on cataloguing such signatures and studying the set they form.

What "new" means here. The novelty is not a new sound law. It is a new *object of study* — the repertoire as a measurable region of a feature space — and a new *order of operations* that keeps the historical apparatus out of the measurement so it can serve as an honest test afterward.

3. Protoform-agnostic by design, not by omission

The question. Every earlier draft of this work apologized for "working without a proto-language," as if it were a limitation to be excused. Is it a limitation — or is it the point?

The intuition. Protoforms are *inferred* objects. They are excellent historical models, but they are not independent observations: they are reconstructed *from* the documented languages using assumptions about how sound change works. If you build your mathematical space *on* protoforms and then ask whether that space confirms the regularities of historical linguistics, you risk circularity — in machine-learning terms, leakage of the target into the inputs. You would be testing the theory against a space the theory helped build.

The inversion, formally. The traditional reconstructive order runs

$$X_{\text{documented}} \longrightarrow \hat{Z}_{\text{proto}} \longrightarrow H_{\text{historical}},$$

where $\hat{Z}_{\text{proto}}$ is a reconstructed ancestor and $H_{\text{historical}}$ the proposed changes. This project runs instead

$$X_{\text{documented}} \longrightarrow \mathcal{M}(X), \quad \text{then, separately,} \quad \mathcal{M}(X) \longleftrightarrow H_{\text{historical}},$$

where $\mathcal{M}(X)$ is a mathematical structure discovered with no protoforms and no precoded laws. The question shifts from "*how did these forms descend from the proto-form we reconstructed?*" to "*what regularities emerge from the documented forms, and how far do they coincide with, diverge from, or extend the regularities historical linguistics proposes?*" That is not a denial of reconstruction; it is a more demanding way to test it.

Not "theory-free." The method is not innocent of assumptions. Transcription, segmentation, concept assignment, the feature matrix, and the alignment algorithm are all measurement instruments, and their influence must be made visible and probed with sensitivity analysis (Chapter 5). The claim is not access to raw data; it is the separation of measurement decisions from the historical hypotheses we later wish to examine. And the discovery phase withholds three things on principle — protoforms, sound laws, and OAS classes — so that each can serve later as an *external* yardstick rather than a hidden input.

The consequence for the whole book. Everything in Parts II–IV is computed under this rule. Part V is where the reserved apparatus (history, and eventually OAS) is allowed back in — as contrast, never as coordinate.

PART II — The instruments (so a reader can reproduce)

4. The data

The question. What exactly goes in, before any mathematics?

The sources. Three, all open and reconstruction-free at the point of use:

- Lexibank — a large aggregation of word lists across families, with every form already segmented into IPA (Segments). This is the primary discovery corpus. We pilot on Indo-European (304 languages available) and Austronesian (978), taking, per family, the languages with the most forms.
- IE-CoR (iecor) — Indo-European with expert cognacy and documented borrowings. We never use its cognacy to *build* the repertoire; we use it only to *validate* the statistical pipeline (Chapter 5) and to run the corpus-regime controls (Chapter 17).
- Glottolog — the genealogical classification, used only to label branches (Chapter 15), never as a computational coordinate.

The units. A concept is a comparison meaning (Concepticon’s ”TOOTH”, ”TWO”...). A form is one language’s word for a concept, given as an IPA segment list. A segment is one IPA sound. That is all the discovery phase consumes.

The caveat about ”family.” In this pilot, ”Indo-European” and ”Austronesian” are *genealogical labels taken as first pilot boundaries*. They are a convenience, not a result; Chapter 18 discusses doing without them. Nothing in Parts II–III depends on the labels being ”correct”; they only delimit which forms are pooled.

Reproducibility. Exact dataset versions, language counts, concept counts, and thresholds are logged in Appendix C; every table in this book names the make target that regenerates it.

5. From form to operator

The question. Given two words for the same concept in two languages, how do we extract *correspondences* without knowing in advance which sounds correspond?

The pipeline, in four steps.

1. Segmentation is already given by Lexibank (IPA Segments).
2. Statistical cognacy. We cluster forms of the same concept into likely cognate sets with LexStat (LingPy), which learns language-pair-specific sound-correspondence scores and clusters by them. No reconstruction, no expert judgment — just recurrent phonetic correspondence.
3. Alignment. Within a cognate set, we align each pair of forms with Needleman–Wunsch, using a substitution cost equal to the fraction of panphon features in which two segments differ. Aligned positions are our candidate correspondences.
4. The feature matrix. Each IPA segment maps, via panphon, to a vector of phonological features (continuant, voice, nasal, coronal, labial, ...). This is the representation on which every later measure is built.

The instruments are not neutral — so we test them. Every step is a measurement choice. Two matter most, and both are audited later: the feature set (Chapter 8 shows results split into basis-dependent vs invariant; Chapter 13 handles panphon’s *ternary* values) and the cognacy threshold (Chapter 5’s validation, and the corpus regimes of Chapter 17, quantify how much statistical cognacy contaminates the result). We find (Chapter 17) that LexStat is *conservative but clean*: at the level of operator types its precision against expert cognacy is near 1.0, even though at the level of individual pairs it is only 0.73.

A worked alignment, by hand. Take ‘three’ in Ancient Greek *treis* and Romanian *trei*. The aligner lays them out to minimize total feature-distance:

```
Greek:   t   r   e   i   s
Romanian: t   r   e   i   -
```

Four columns match (t,r,e,i); the last is a *gap* (Greek has a final s, Romanian does not). Gaps are not operators — they are insertions/losses, recorded separately. Now Bulgarian *tri* vs Yiddish *draj*:

```
Bulgarian: t   r   i
Yiddish:    d   r   a   j
```

Column 1 gives $t \ d = \{\text{voi}\}$ (our familiar atom); column 3 gives a vowel correspondence $i \ a$; the final j is again a gap. Every aligned, non-identical, non-gap column is one operator instance. Run this over all cognate pairs and you have counted the whole repertoire — mechanically, with no reconstruction anywhere.

Reproduce. `make family FAMILY="Indo-European"` runs steps 1–4 and writes the correspondence table; `make cognate-eval` runs the validation against IE-CoR.

6. The operator $\Delta(a, b)$

The question. What, precisely, is an “operator” in this book — and what is it *not*?

The definition. For two aligned segments a, b , the operator is the set of primary features in which they differ:

$$\Delta(a, b) = \{ k : \phi_k(a) \neq \phi_k(b) \},$$

where ϕ is the panphon feature map. Equivalently, over binary features, $\Delta(a, b) = \phi(a) \oplus \phi(b)$, the XOR of the two feature vectors. We keep operators of one to three features (single changes and small bundles), which are the phonologically interpretable ones.

(b/d) Worked reading. $d \quad z$ differ in continuancy and stridency $\Rightarrow \{\text{cont} + \text{strid}\}$ (a spirantization signature). $t \quad s \Rightarrow \{\text{cont} + \text{strid}\}$; but $t \quad f \Rightarrow \{\text{cont} + \text{cor} + \text{strid} + \text{lab}\}$ — *not the same operator*, because f is a labiodental and differs from t in place as well. The feature representation makes the distinction automatically; lumping both as “lenition” would erase it.

What the operator is NOT. It is symmetric: $\Delta(a, b) = \Delta(b, a)$. It does not distinguish $t \rightarrow s$ from $s \rightarrow t$. It is a *differential signature*, a contrast — not a directed historical change. The frequently-repeated fact that “each operator is its own inverse” ($\Delta \oplus \Delta = \emptyset$) is a *consequence of this symmetry*, not a phonological discovery. Directed history, when we want it, needs a richer object,

$$T = (a, b, c, p, \ell_1, \ell_2, w) \quad \text{with signature} \quad \sigma(T) = \Delta(a, b),$$

carrying source, target, context c , position p , the two languages, and a weight — so that $T_{t \rightarrow s} \neq T_{s \rightarrow t}$ even though their signatures coincide. That object belongs to Part V; Parts II–IV study the symmetric signatures.

Reproduce. The operator of any segment pair is `delta(a,b)` in `src/algebra.py`; the family’s operator set is what make `algebra FAMILY="X"` reads.

6.1 The same operators live *inside* a single language

(*Illustrative interlude. The examples in this section are drawn from well-known grammatical descriptions, not computed from our reconstruction-free corpus. They are here to teach one thing: the operators the method finds between cognate languages are the same objects that alternate inside a single grammar — in plurals, verb paradigms, and derived words. Sound-difference is sound-difference, whether it sits between Greek and Armenian or between two forms of one English word.*)

Our operators compare two *languages*. But the very same feature-differences show up when a *single* language builds or inflects a word. Watch `{voi}` — the one-feature voicing operator from Chapter 0 — appear again and again, now *within* one grammar:

language	construction	alternation	operator
English	plural of <i>wife</i> → <i>wives</i>	f v	<code>{voi}</code> (+cont, strid)
English	noun <i>house</i> hæʊs → verb <i>house</i> hæʊz	s z	<code>{voi}</code>
German	<i>Tag</i> tɑ:k → plural <i>Tage</i> tɑ:gə (final devoicing)	k g	<code>{voi}</code>
Dutch	<i>huis</i> hæys → plural <i>huizen</i> hæyzə	s z	<code>{voi}</code>

The plural of *wife*, the verb *to house*, the German declension — three unrelated pieces of grammar, one and the same operator `{voi}`. This is why the repertoire is not an artifact of comparing languages: it catalogues feature-differences that grammar itself uses productively.

Two-feature operators appear inside grammars too, and so do the famous “corridors” of Chapter 12:

- A verb paradigm. English *was* / *were* is a single verb’s own past tense: s → r. Historically this is Verner’s Law followed by rhotacism (s → z → r); as an *operator corridor* it is `{voi}` composed with a sonorant step — the very s → z → r path the method surfaces from cognates in Chapter 12. A learner meets the corridor twice: across the family, and inside one irregular verb.
- Latin declension. *flōs* ‘flower’ → genitive *flōr-is*: s → r, the same rhotacism corridor, now a case ending.
- Vowel ablaut in verbs. English *sing* / *sang* / *sung*, or Sanskrit *vid-* / *véd-a*: the vowel operators (`{hi}`, `{back}`, `{lo}`) that Chapter 13 treats as *graded* dimensions are exactly what conjugation exploits to mark tense.

The teaching point. An operator is a unit of *phonological difference*. The method reads it off cognate pairs because that isolates it cleanly; but the same units are the working parts of every plural, tense, and derivation. When Part V speaks of adding a *directed, context-conditioned* layer (who changes into what, and when), morphology is exactly where those directed, conditioned rules become visible inside one grammar. (*None of the numbers in this book use these examples; they are the intuition, not the measurement.*)

PART III — The repertoire and its geometry

This is the heart of the book. Every number below was computed under the standing rule (no proto-forms, no laws, no OAS) and is reproducible with the named make target. We report Indo-European (IE) and Austronesian (AN) side by side.

7. Repertoire O vs generated subspace $\langle O \rangle$

The question. We have a family's set of operators, O . What is the *space* it lives in, and how much of that space does it actually use?

Two objects, kept separate. The repertoire O is the set of observed operators. Because operators are vectors over $\mathbb{F}_2$ (features under XOR), O *generates* a linear subspace

$$\langle O \rangle = \text{span}_{\mathbb{F}_2}(O),$$

of dimension $r = \dim \langle O \rangle$ (the rank, Chapter 8). Crucially, O is not a subspace: it is not closed under XOR and does not contain the zero vector. O is a *region*; $\langle O \rangle$ is the *capacity* it implies. The gap between them,

$$\langle O \rangle \setminus O,$$

is the set of operators the repertoire *could* generate but does not attest — the holes (Chapter 12).

The two families.

	$ O $ (types)	rank r	$ \langle O \rangle - 1 = 2^r - 1$
Indo-European	67	12	4095
Austronesian	22	10	1023

Why this separation matters. Almost every interesting question is really about the *relationship* between O and $\langle O \rangle$: how much of the capacity is used (Chapter 9), whether combining elements of O stays in O (Chapter 10), and which of the ~ 4000 generable IE operators are the ones actually avoided (Chapter 12). Confusing the two — treating the repertoire as if it were the subspace — was the central error the earliest drafts made, and the reason the “do they form a group?” question was the wrong one to ask.

Limit. $\langle O \rangle$ is an *abstraction*: many of its $2^r - 1$ elements are not differences that could occur between two real segments. Chapter 9 replaces this too-large denominator with realizable universes.

Reproduce. make universes FAMILY="X".

8. How many independent “knobs” does a family really use?

(A note on the two families used throughout. Indo-European, the familiar family that includes English, Spanish, Russian, Greek, Armenian and Hindi; and Austronesian, a very large family of roughly a thousand languages spread across the Pacific and island Southeast Asia — Malagasy, Malay, Tagalog, Fijian, Māori, Hawaiian. We compare the two throughout.)

The puzzle this chapter answers

We will meet a statement that sounds contradictory: “Indo-European changes along 8 atoms, yet its space has rank 12.” How can there be more independent directions (12) than single-feature changes (8)? Clearing that up teaches the two ideas we need — rank and corank — from scratch.

Atoms and molecules

Recall an operator is *the set of features that differ* between two sounds. An atom changes exactly one feature; a molecule changes several at once.

- Atom: {voi} (voicing alone) — seen in d t, g k, b p, s z.
- Molecule: {cont+strid} (two features) — seen in d z, t s.

Indo-European uses 8 kinds of atom; Austronesian only 3. So far, so simple.

Rank: how many *independent directions* of change

Think of each phonological feature as a knob you can flip. A family's operators are the *combinations of knob-flips* it actually performs. The rank asks: how many knobs would you minimally need — as independent controls — to reproduce every combination the family uses?

Here is the subtlety, and the resolution of the puzzle. A feature can be a genuine independent direction *even if the family never flips it on its own*. Nasality (nas), labiality (lab) and rounding (round) never appear as Indo-European atoms — they only ever show up *inside* molecules like {nas+son+voi}. But they are still independent knobs: no combination of the other features reproduces them. So they count toward the rank even though they are not atoms.

That is the whole answer: atoms are the changes the family makes with a single knob; the rank counts every independent knob that moves at all, alone or in company. Indo-European makes single-knob changes on 8 features but *moves* 12 independent knobs in total — hence 8 atoms, rank 12.

A tiny worked instance. Suppose a family used just three operators: {voi}, {cont}, and {voi+cont}. It has 2 atoms ({voi}, {cont}) and rank 2 — because the third operator is just the first two flipped together ({voi} $\oplus$ {cont} = {voi+cont}), no new direction. Now add {nas+voi}: still only {voi} and {cont} as atoms (2 atoms), but rank rises to 3, because nas is a new independent knob even though it never appeared alone. That is exactly how IE reaches rank 12 with 8 atoms.

Corank: how many knobs turned out to be *redundant*

Count the features that move at all; call it n . Count the independent directions; that is the rank r . The difference $n - r$ is the corank — the number of knobs that are not independent, i.e. that always move as a fixed combination of others.

family	features that ever change (n)	rank r	corank $n - r$
Indo-European	12	12	0 — every moving knob is independent
Austronesian	11	10	1 — one knob is redundant

Indo-European has corank 0: all 12 knobs it moves are genuinely independent. Austronesian has corank 1: of the 11 knobs it moves, only 10 are independent; one is pinned to the others.

Which knob is redundant in Austronesian — and does it mean anything?

We can name the redundancy exactly. Across Austronesian's operators, whenever you know how two knobs — cor (coronal, made with the tongue tip/blade) and lab (labial, made with the lips) — are

set, the third knob *ant* (anterior, front-of-mouth) is already determined: in this sample, $\text{ant} = \text{cor} \oplus \text{lab}$. It never varies independently of those two.

Be careful what this does and does not say. It does not say anteriority “is really” coronal-plus-labial in some deep sense. It says only that, *in the specific set of correspondences Austronesian happens to use*, the anterior knob never moves on its own — its pattern coincides with a combination of the other two. It is a fact about this family’s *choices*, phrased in the language of linear algebra.

Is the redundancy real, or an accident of our feature chart?

This is the crucial control, and it is easy to state. Maybe $\text{ant} = \text{cor} \oplus \text{lab}$ is baked into the panphon feature chart itself — true of *any* set of sounds, not special to Austronesian. To check, we compute the rank not on the operators Austronesian *uses*, but on every difference that its sound inventory could possibly produce — all pairwise differences among the 88 consonants Austronesian actually has. Call that the *realizable universe* U_S . Result: U_S has rank 12, corank 0 — the feature chart *would allow* all 12 knobs to be independent. The redundancy is therefore not in the chart; it appears only in the operators Austronesian selects. So $\text{ant} = \text{cor} \oplus \text{lab}$ is a genuine property of the *system’s choices*, not an artifact of how we wrote the features. (Run it: `make repr-control FAMILY="Austronesian".`)

Which of these numbers are “real,” and which depend on our bookkeeping

A last honest point. Some numbers change if we rename or recombine the features; others do not.

- Depends on the feature chart (bookkeeping): the *atom count* (8 vs 3), which features happen to appear alone, the size of each operator. An “atom” is only atomic *relative to the feature chart we chose*.
- Does not depend on it (intrinsic): the rank, the corank, the redundancy relations, and the circuit structure of Chapter 11. Rewrite the features in any invertible way and these stay the same.

So when we compare families, we lean on the *invariant* quantities (rank, corank, circuits) and treat the atom count as a readable-but-chart-dependent summary — never as an intrinsic dimension. It is meaningful only because our features carry real articulatory content, not arbitrary labels.

Reproduce. `make algebra FAMILY="X" (atoms, rank, corank); make repr-control FAMILY="X" (the control).`

8.1 The correspondence graph — the repertoire, drawn

The most direct picture of a repertoire is a graph whose nodes are sounds and whose edges are correspondences, one edge for each recurrent $a \leftrightarrow b$, coloured by the *kind* of change (voicing, fricativization, palatalization, place, prenasalization) and thickened by frequency. This is, strictly, a *segmental graph of correspondences labelled by operator* (Chapter 21 distinguishes it from the operator graph G_O and the language graph G_L). Read it as the “state space” of consonant correspondence for each family.

The Indo-European graph is denser and reaches more places of articulation; the Austronesian one is sparser and organized around voicing and its prenasalized bundle $t^n d, k^n g, p^n b$. Neither picture required a single reconstruction.

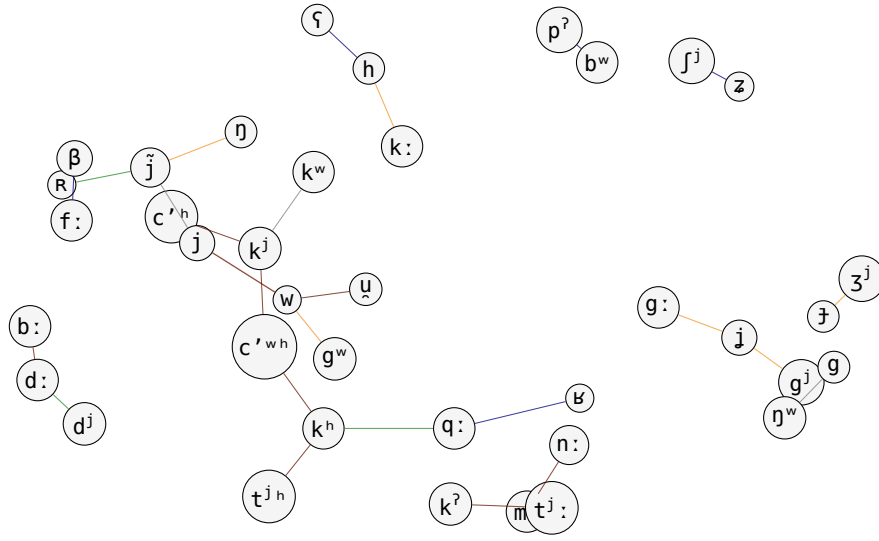


Figure 1: Segmental correspondence graph G_S of Indo-European. Nodes are segments; edges are recurrent correspondences, with thickness $\propto$ frequency and colour by change type: **voicing**, **fricativization/lenition**, **palatalization/height**, **place**, **prenasalization**.

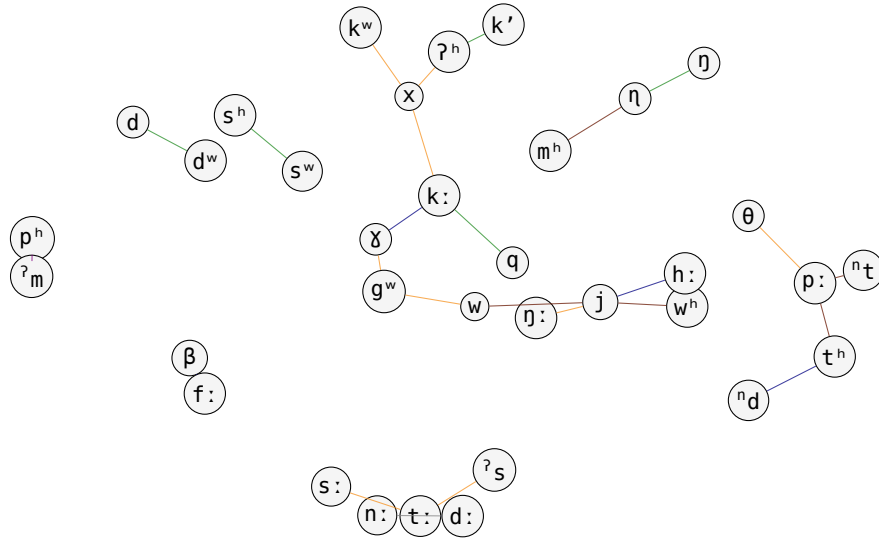


Figure 2: Segmental correspondence graph G_S of Austronesian. Nodes are segments; edges are recurrent correspondences, with thickness $\propto$ frequency and colour by change type: **voicing**, **fricativization/lenition**, **palatalization/height**, **place**, **prenasalization**.

9. Occupancy in three universes

The question. IE uses 67 of ~ 4095 generable operators — 1.6%. Are phonological systems really that restrictive?

Worked example — ‘fire’. One word already shows the logic. Indo-European ‘fire’ across seven languages — Bulgarian *ogən*, Romanian *fok*, Serbo-Croatian *vatra*, Seychelles Creole *dife*, Albanian *zjar*ⁿ, Eastern Armenian *kʰrakʰ*, Western Armenian *grag* — contributes only a handful of operators, among them our atom $\{\text{voi}\}$ (from Armenian *kʰ g*, *kʰrakʰ* vs *grag*). Each word adds a few operators; pooled over the whole vocabulary they converge on just 67 distinct types. Sixty-seven out of an algebraic 4095 sounds like almost nothing — until you ask the right denominator (below).

The intuition. No — and this is one of the most important corrections in the book. The 1.6% compares the repertoire to the full *algebraic* span, most of which consists of feature-bundles that could never be a difference between two real segments, or that never had the chance to occur in the corpus. The honest question is “of what was actually available, how much is used?” — and that needs a smaller, realistic denominator.

Three nested universes.

$$O \subseteq \Omega_D \subseteq U_S \subseteq \langle O \rangle,$$

where U_S is realizable (differences between attested segments), and Ω_D is the opportunity universe (differences that actually occur in aligned positions of the corpus, at any support). Three occupancies follow:

$$\rho_{\text{alg}} = \frac{|O|}{2^r - 1}, \quad \rho_{\text{seg}} = \frac{|O|}{|U_S \cap \langle O \rangle|}, \quad \rho_{\text{opp}} = \frac{|O|}{|\Omega_D \cap \langle O \rangle|}.$$

The result.

	ρ_{alg}	ρ_{seg}	ρ_{opp}
Indo-European	0.016	0.48	0.63
Austronesian	0.022	0.43	0.48

Of what the corpus made available, Indo-European uses 63% and Austronesian 48%. The “98% unused” was almost entirely the *algebraic* abstraction being enormous — not languages being restrictive. The repertoire is, in the only sense that matters phonologically, fairly dense.

Limit. Ω_D depends on the corpus and the aligner; it is an *opportunity* estimate, not a phonological law. But it is far closer to the phenomenal reality than $2^r - 1$, and the three numbers *together* separate three kinds of constraint: algebraic, representational, and language-specific.

Reproduce. make universes FAMILY="X".

10. Additive structure — the mathematical heart

The question. When we compose two operators — XOR their feature-differences — does the result tend to be another *observed* operator? Is the repertoire *additively organized*, or just an arbitrary set?

The measure and its conditioning. The composition-realization index

$$C(O) = \frac{|\{\{u, v\} \subseteq O : u \oplus v \in O\}|}{\binom{|O|}{2}}$$

(for $u \neq v$ in $\mathbb{F}_2$, $u \oplus v \neq \emptyset$ automatically, so the denominator is just $\binom{|O|}{2}$). Observed: IE 0.220, AN 0.234. Conditioned on realizability, $C_\Omega(O)$ asks: *of the compositions that land in the opportunity universe, how many are observed?* — IE 0.81, AN 0.66.

A worked composition. $\mathbf{b} \ \mathbf{p} = \{\text{voi}\}$ and $\mathbf{p} \ \mathbf{f} = \{\text{cont} + \text{strid}\}$ compose to $\{\text{voi} + \text{cont} + \text{strid}\}$; the pair $\mathbf{b} \ \mathbf{f}$, when it occurs, necessarily carries that signature — the algebra *predicts* which operator it would be, and we can check whether the system realizes it.

Does it beat chance? A hierarchy of nulls. A single number is descriptive until compared to a null. We built six nested nulls, each preserving more structure, and asked whether observed $C(O)$ exceeds them (Z -scores; empirical p_{MC} from 500 simulations):

null (preserves...)	IE Z	AN Z
0 size + weight ≤ 3	+15.7	+12.0
1 Hamming weights	+14.8	+9.0
2 exact margins (swap-MCMC)	+7.5	+3.6
3 rank / span $\langle O \rangle$	+45.6	+13.7
4 realizable U_S	+9.2	+7.3
5 opportunity Ω_D	+6.3	+5.9

All six survive at $p_{MC} \leq 0.004$. Even when we draw $|O|$ operators from *exactly the differences the corpus made available*, the observed repertoire is significantly more composition-closed. Additive structure is a robust, size-controlled property. (Bootstrap CIs: IE [0.214, 0.235] tight; AN [0.166, 0.217], so AN's $C(O)$ is more support-sensitive.)

Three converging signatures. Against the Ω_D null: triple density τ is high (IE $Z = +6.6$, AN $+6.5$), the doubling constant $\kappa = |O \oplus O|/|O|$ is *low* (IE $Z = -5.1$, AN -6.0 ; small doubling = strong additive structure), and additive energy E is high (IE $Z = +7.4$, AN $+7.3$). The repertoire behaves like an approximate subspace, possibly an affine one $v+W$ (a family of operators sharing an obligatory feature).

The honest caveat — this is not (yet) a genealogical signal. At the level of operator *types*, this additive structure is largely a property of the phonological inventory and feature representation, not of history. The evidence is decisive: when we permute concept labels to destroy all real cognacy and semantics (regime D_R , Chapter 17), $C(O)$ and τ stay *as high or higher* than under expert cognacy D_G . So $C(O)$ exceeds *size-matched random* (the nulls above) — that is real — but it does not distinguish genealogy from inventory. Where genealogical signal lives is the subject of Chapter 14 and Part IV: in the *distributions*, not the type sets.

Reproduce. make nulls FAMILY="X" (the hierarchy); make additive FAMILY="X" (τ, κ, E).

11. Circuits and binary matroids

The question. Can we describe the repertoire's dependency structure in a way that does *not* depend on the arbitrary choice of feature basis?

The matroid. The operator $\times$ feature incidence matrix defines a binary matroid. Its most interpretable invariants are the circuits: minimal sets of operators whose XOR is zero. A size-3 circuit $u \oplus v \oplus w = 0$ means $w = u \oplus v$ — three operators locked in an additive triangle.

Worked example — a real circuit. Three Indo-European operators are locked in an additive triangle: $\{\text{ant}\}$ (the sibilant shift $s \rightarrow \text{ʃ}$), $\{\text{strid}\}$ ($s \rightarrow \theta$), and $\{\text{ant+strid}\}$ ($\text{ʃ} \rightarrow \theta$). Check it: $\{\text{ant}\} \oplus \{\text{strid}\} = \{\text{ant+strid}\}$, i.e. $s \rightarrow \text{ʃ}$ composed with $s \rightarrow \theta$ gives $\text{ʃ} \rightarrow \theta$ — and all three are operators the family actually uses. That closed triangle is a size-3 circuit. It does not depend on how we named the features: rename or recombine them and the triangle is still there. Another, involving the hub: $\{\text{voi}\} (d \rightarrow t) \oplus \{\text{ant}\} (s \rightarrow \text{ʃ}) = \{\text{ant+voi}\} (z \rightarrow \text{ʃ})$.

The result. IE has 162 size-3 circuits, AN 18. In *both* families the operator participating in the most circuits is **{voi}** — voicing is the additive hub of the repertoire. Because circuits are a matroid in-

variant, this is a basis-independent fact: it does not depend on how we named or coordinatized the features.

Why it matters. Two systems could use different feature labels yet share the same matroid — the same pattern of dependencies. The matroid is therefore the right object for asking “*do two systems have the same dependency structure?*”, a question much closer to the goal of finding patterns more abstract than any particular notation. (Part V returns to this comparison.)

Reproduce. make additive FAMILY="X".

12. The span as a linear code

The question. We said the repertoire uses little of its span. *Which* generable operators are missing, and how far are they from what is present?

Worked example — ‘star’. Indo-European ‘star’ (Bulgarian *zvezda*, Ancient Greek *aster*, Serbo-Croatian *zvezda*, Albanian *ül*, Judeo-Tat *astara*) yields the correspondences $s\ z = \{\text{voi}\}$, $r\ l = \{\text{hi}+\text{lo}\}$, and $v\ s = \{\text{cor}+\text{lab}+\text{voi}\}$, all of which are *observed* operators. Now ask about a nearby operator the family could generate but this word did not use — say $\{\text{voi}+\text{strid}\}$ (voicing plus stridency together, as in a bare $z\ s$ without the extra place features). If $\{\text{voi}\}$ is present and $\{\text{strid}\}$ is present but $\{\text{voi}+\text{strid}\}$ is not, that missing combination is a hole at distance 1: available in the corpus’s opportunities, one feature away from what is attested, yet unchosen. Cataloguing exactly these near-misses is the geometry below.

The code and the holes. Treat $\langle O \rangle$ as a binary linear code. Among the operator-shaped vectors (weight ≤ 3) that the code generates, some are observed (O) and the rest are holes. For each hole x we measure its depth $d(x, O) = \min_{o \in O} d_H(x, o)$ (Hamming distance to the nearest observed operator).

The geometry of holes.

	weight- ≤ 3 generable	observed	holes	at $d = 1$	holes in Ω_D
Indo-European	298	67	231	160	40
Austronesian	119	22	97	53	24

Most holes sit at $d = 1$ — one feature away from completing a pattern. The linguistically loaded ones are the holes that lie in Ω_D (40 in IE, 24 in AN): operators that were *available* in the corpus yet went *unchosen*. These are the concrete realization of $\langle O \rangle \setminus O$ — the “generable-but-avoided” object — and the natural candidates for interpretation: absences that are choices, not impossibilities.

Limit. For IE the code is the whole space (rank 12 = 12 features), so “the code” is only a proper, informative object where the corank is positive (AN); the *hole geometry*, however, is informative for both, because it is measured on the operator-shaped (weight ≤ 3) slice, not the whole span.

Reproduce. make additive FAMILY="X".

13. Non-binary dimensions (the “XOR-break”)

The question. The composition law $\Delta(a, c) = \Delta(a, b) \oplus \Delta(b, c)$ is an identity for binary features. Yet on real data it holds “only” 98.1% of the time in IE. If it is an identity, how can it ever fail?

The diagnosis. It cannot fail for genuinely binary features. It fails precisely where a feature is not binary: panphon features are *ternary* (+1, 0, −1). All five IE failures are triples where one feature takes three distinct values across a, b, c — every one of them in height (hi): $\text{ɫ} \quad \text{ɭ} \quad \text{r}, \text{ɫ} \quad \text{r} \quad \text{ʌ}$,

... So the "XOR-break" is not a violated law; it is a diagnostic that flags which dimensions of the system are *multivalued* and cannot be modeled with a single bit.

What it does and does not show. It localizes a *multivalued dimension*; it does not by itself prove a totally-ordered phonetic scale. ɭ , ɭ^j , r , ɰ vary at once in laterality, roticity, and place; whether the right geometry is a total order, a partial order, or a state graph is a separate question. Austronesian shows no XOR-break in its 47 relevant triples — which means *none was detected in this sample* (with $n = 47$ a small true rate could be missed), not that AN "has no graded dimensions."

The fix, when wanted. For a ternary feature one can use a one-hot encoding $-1 \mapsto e_1, 0 \mapsto e_2, +1 \mapsto e_3$, restoring exact composition for *categorical* states; or, for *ordered* states, a path metric with $d(-1, +1) = 2$. The general system is then a product of typed spaces

$$\mathcal{F} = \prod_{j \in B} \mathbb{F}_2 \times \prod_{j \in C} \Delta_{m_j} \times \prod_{j \in O} \mathbb{Z},$$

binary $\times$ categorical $\times$ ordered — or, most generally, a box product of per-feature state graphs. The mathematics adapts to the feature instead of forcing every feature to be a bit.

Reproduce. `make algebra FAMILY="X"` (reports the break and the graded features).

14. Distributions, not only types

The question. The repertoire records *existence*. But how much of the structure is in the *frequencies* and in *what each operator co-occurs with*?

Worked example — 'nose' vs the voicing atom. Austronesian 'nose' is *ihu* in Hawaiian, Maori, Tongan, Tuamotuan, Rapa Nui, Marquesan — *identical everywhere*. Its correspondence set is empty: zero operators, a word so conserved it contributes nothing to the repertoire. At the other extreme, $\{\text{voi}\}$ (d t , b p , s z , ...) recurs tens of thousands of times across the vocabulary. The repertoire is not a flat list: a few operators carry almost all the mass, many carry almost none, and some words (like *ihu*) carry none at all. That imbalance is exactly what entropy measures.

Distribution and its concentration. With per-instance counts we form $P_L(o)$, its Shannon entropy H , and the effective number of operators $N_{\text{eff}} = 2^H$. IE: $N_{\text{eff}} = 24$ of 119 types; AN: 18 of 63. A handful of operators dominate each family.

Three geometries. *Unweighted* (what exists), *frequency-weighted* (how often), *confidence-weighted* (how securely inferred, using alignment cost as a proxy). The top-8 operators coincide 7/8 between frequency and confidence in both families — the repertoire's ranking is robust to confidence weighting. (Support, distribution, and confidence are nonetheless different robustnesses, and should be reported separately.)

Mutual information — where dependence lives. $I(O; \cdot)$ (normalized) between the operator and four variables:

	branch R	concept C	position	context Γ
Indo-European	0.14	0.16	0.06	0.05
Austronesian	0.22	0.41	0.06	0.09

Dependence is strongest on concept (partly a confound: the concept fixes which segments are in play), then on branch — and here is the key: branch information is *higher in Austronesian* (0.22) than in Indo-European (0.14), consistent with the branch analysis of Part IV. Dependence on position and coarse context is *low*.

The synthesis of Part III. The additive/geometric structure of the *type set* is largely a property of the phonological representation, shared across the two families and even reproduced by concept-permuted artifacts. The genealogical signal — the thing historical linguistics cares about — is *second-order*: it lives in the distributions (this chapter's branch information), not in the type sets or their additive closure. That is why the fine, context-conditioned analysis (Part V's tensor) is the right next instrument; coarse context shows little here.

Reproduce. make distributions FAMILY="X".

PART IV — Systems and comparison

Comparative linguistics compares *forms* and reconstructs *trees*. This part asks a different comparative question: do the repertoires of related systems overlap more than chance, and where does any genealogical signal actually reside? The answers are more sobering — and more interesting — than the enthusiasm of a first look.

15. The family as a union of branches

The question. Is the "algebra of Indo-European" a single thing, or the superposition of several branch algebras dominated by whichever branches are best sampled?

Worked example — 'fire', within a branch vs across. Inside the *Armenic* branch, Eastern kʰrakʰ and Western grag differ by a single clean operator, kʰ g = {voi} — a tidy, intra-branch correspondence. Across *different* branches, 'fire' barely aligns at all: Bulgarian ogən (Slavic), Romanian fok (Romance), Albanian zjarʰ share almost no material. So the intuition "branches share their own operators" seems obvious. The surprise (below) is that, *measured against a proper null*, it is only true for Austronesian — not for Indo-European.

The clean decomposition. Partition each family into its genealogical branches (Glottolog, cut where clades reach a manageable size), run the whole pipeline *within* each branch, and take the family repertoire as the union $O_F = \bigcup_r O_r$. For each operator define its persistence $p(o) = \frac{1}{R} \# \{r : o \in O_r\}$ and the histogram $H_k = \#\{o : o \text{ in exactly } k \text{ branches}\}$, which by construction sums to $|O_F|$. Three nuclei, named unambiguously: $K_\cap$ (in *all* branches), $K_{1/2}$ (in $\geq \lceil R/2 \rceil$), K_α (general threshold).

The decomposition. IE (7 branches): $|O_F| = 97$, $\Sigma H_k = 97$ [7], $K_\cap = 12$ (the atoms {ant} {cont} {voi} are in every branch), $K_{1/2} = 38$, exclusives = 26. AN (11 branches): $|O_F| = 48$, $K_\cap = 4$, $K_{1/2} = 18$, exclusives = 15. (*This corrects a v3 table that mixed two universes and appeared to double-count; the fix was to use $O_F = \bigcup O_r$ consistently and let $\Sigma H_k = |O_F|$.*)

The decisive control. A raw Jaccard between branch repertoires (IE 0.42, AN 0.51) means *nothing* by itself. We compare it to a grouping null: partition the same languages into arbitrary groups of the same sizes and recompute. The result overturns the earlier "superposition" story:

	mean Jaccard (real)	arbitrary group- ings	Z
Indo-European	0.42	0.45 ± 0.05	−0.7 (indistinguishable)
Austronesian	0.51	0.57 ± 0.01	−5.4 (real branches more differentiated)

So at the level of operator types, Indo-European branches overlap *no more than random language groupings* — there is no type-level genealogical signal in IE; the shared core is shared *inventory*. Austronesian branches, by contrast, are significantly more differentiated than chance (Formosan vs Oceanic have very different inventories) — a genuine signal.

Limits. Under rarefaction to equal branch sizes, the nucleus size is unstable (IE $K_{1/2}$ falls from 42 to ~15), so branch *richness* counts are sampling artifacts and must not be interpreted. The robust facts are: (i) the union/histogram decomposition; (ii) the grouping-null result (no IE type-signal; AN signal). The genealogical signal, where it exists, is distributional (Chapter 14), not in the type sets.

Reproduce. make superposition FAMILY="X"; add TF_RAREFY=3 TF_NULLBRANCH=3 for the controls.

16. Negative controls

The question. How do we know a "system" pattern is not just general phonetic geometry?

(c/e) The controls, and their status.

- Concept permutation (D_R , Chapter 17): destroys form $\leftrightarrow$ concept links. *Run* — and it matches the real additive structure, showing much of it is inventory-driven.
- Branch-label permutation (Chapter 15): arbitrary groupings of the same sizes. *Run* — the grouping null above.
- Pseudo-families / inventory controls: random language groups matched on size, segment inventory, concept coverage, alignment count. *Partially run* via the grouping null; a fuller inventory-matched control is scheduled.
- Geographic control: compare genealogical groups with areal groups of the same size. *Planned*.
- Temporal control: the corpus mixes ancient and modern states; the absence of protoforms does *not* force us to drop time — we can use documented dates. Build a temporal network $G(t)$ or attach a date interval to each form and study repertoire change without reconstructing unattested states. *Planned*.

The lesson. A number is a finding only relative to the strictest control it survives. The additive structure survives size-matched random draws (Chapter 10) but not the concept-permutation artifact at the type level (Chapter 17) — both statements are true and must travel together.

Reproduce. `make regimes` (concept permutation); `make superposition ... TF_NULLBRANCH=3` (grouping).

17. Four corpus regimes: D_G, D_L, D_C, D_R

The question. Rather than discard the concept-only corpus as "dangerous," can we use it — and its degradations — as a control that separates *sources* of structure?

Four regimes on IE-CoR. D_G expert cognacy · D_L statistical (LexStat) cognacy · D_C same-concept, no cognacy · D_R concepts permuted (the artifact floor).

The fingerprints.

regime	$ O $	$C(O)$	τ	κ
D_G expert cognacy	65	0.169	0.166	14.4
D_L LexStat	44	0.162	0.158	12.3
D_C concept-only	108	0.204	0.202	14.6
D_R concept-permuted	104	0.215	0.212	14.6

The honest reading. The artifact D_R has additive structure *as high as or higher than* expert cognacy D_G . Therefore the additive organization of the operator type set is largely a property of the segment inventory and its frequencies — not of genealogy, and not even of conceptual equivalence ($D_C \approx D_R$). This is the type-level counterpart to Chapter 15's grouping-null result. It also tells us where the LexStat validation matters: at the type level LexStat's precision against D_G is ≈ 1.0 (it recovers real operators cleanly), even though at the pair level its precision is only 0.73 — the type inventory absorbs individual cognacy errors.

Loans as system-influence, not annotation (idea 2). The right way to ask "cognate vs loan, mathematically" is not to define it but to *observe* whether loans move the repertoire: compare O with and without loan-flagged forms — do loans add operators, or fall inside the existing O ? The machinery is implemented; the current data cannot answer it (IE-CoR marks borrowings only in free-text comments; Lexibank's Loan column is empty for IE and AN). This needs a loan-annotated corpus such as WOLD, and is a clean next experiment.

Reproduce. make regimes.

18. Family-blind: when yes, when no

The question. Both "systems" here were defined by genealogical labels. Could we compare systems *without* presupposing families?

The intuition, and the caution. Yes — assign each language (or language pair) an operator distribution $P_{ij}(o)$, define a distance between distributions, cluster, and see what systems *emerge*. But this points at the universal/typological questions we are deliberately not asking in this phase. Family-blind analysis belongs to the *historical-holdout* programme (Chapter 20, phase C), where it is a powerful discovery tool — not to the present intra-system study, where imposing it would blur exactly the boundaries we are studying.

The right distance. Not Jaccard (which treats two near-identical operators as wholly different) but a Wasserstein distance over operator distributions, with ground cost $d(o_1, o_2)$ from Hamming or articulatory distance — so two systems using *neighboring* operators are close even if their type sets do not coincide.

Why not now. Family-blind clustering answers "what large-scale systems exist?"; this book answers "what is the structure *within* a given system?" Mixing them would trade a sharp question for a diffuse one. We flag the method and its tools, and reserve it.

Reproduce. Not yet a target — specified here as a Part-V experiment.

PART V — Toward history and toward AI

Everything so far kept history out of the computation. This part is where the reserved apparatus is allowed back — as *contrast*, never as coordinate — and where the programme’s larger ambitions are laid out. These chapters are a research programme, not finished results.

19. Letting historical laws emerge without inserting them

A sound law is rarely a single operator; it is a *bundle* of correspondences, tied to particular languages and often conditioned by context. So we ascend in orders. Order 0 is this book: the aggregate repertoire O_L . Order 1 builds the tensor X_{ijo} = normalized frequency of operator o between languages i, j ; a non-negative factorization $X \approx \sum_q \lambda_q a_q \otimes a_q \otimes b_q$ yields components where a_q selects a set of languages and b_q a *bundle* of operators — a recurrent transformation cluster the algorithm was never told the name of. Order 2 adds context X_{ijoc} (preceding/following segment, position, stress, morphology) and seeks $P(o \mid c, i, j)$. Here Minimum Description Length gives an emergence criterion: a regularity earns the name *rule* when a small set of context-conditioned operators compresses a large slice of the corpus, i.e. minimizes $L(\mathcal{R}) + L(D \mid \mathcal{R})$. "Grimm’s law" would be *discovered* as a compressive bundle over Germanic, not supplied in advance.

20. A historically-blind validation protocol (historical holdout)

Phase A — discovery: no protoforms, laws, change classes, direction; in some runs, not even family labels. Phase B — mathematical stability: bootstrap over languages and concepts; robustness to the feature matrix, the cognacy threshold, and the aligner; superiority over the null hierarchy. Phase C — blind system discovery: cluster operator profiles with no genealogical labels; ask whether families, areas, or continua emerge, and which languages land in surprising places. Phase D — external contrast: only now compare with reconstructions, laws, chronologies, genealogies, and documented contact. A non-match is itself a result — it localizes either a corpus limit, a representational gap, or a place where traditional terminology lumped mathematically heterogeneous phenomena. This design is what protects the project from the charge of merely re-encoding what historical linguistics already handed it.

21. A general architecture

Four layers plus an external one. Phenomenal: $D = \{(\ell, t, g, c, w, \pi(w))\}$ (language, date, place, concept, form, phonetics). Relational: weighted alignments between forms. Operatory: the signatures $\Delta(a, b)$ and the distributions $P_{ij}(o)$. Emergent-structural: $\mathcal{A}_L = (O_L, V_L, G_S, G_O, G_L, P_L)$ — repertoire, span, and three graphs. External-historical: reconstructions, laws, genealogies, contacts — used only in $\text{Compare}(\mathcal{A}_L, H)$. The three graphs must be kept distinct: G_S segmental (nodes = segments, edges = correspondences — what the current figures actually show), G_O of operators (nodes = operators, edges = composition $u \oplus v \in O$), and G_L of languages (nodes = languages, edge weight = distance between operator distributions). History does not disappear in this architecture; it changes epistemic position, from initial coordinate to final contrast.

The segmental graph G_S was shown in §8.1. Here are the other two. First the operator graph G_O , whose edges are additive closure ($u \oplus v \in O$) — the visual face of the composition index $C(O)$ of Chapter 10:

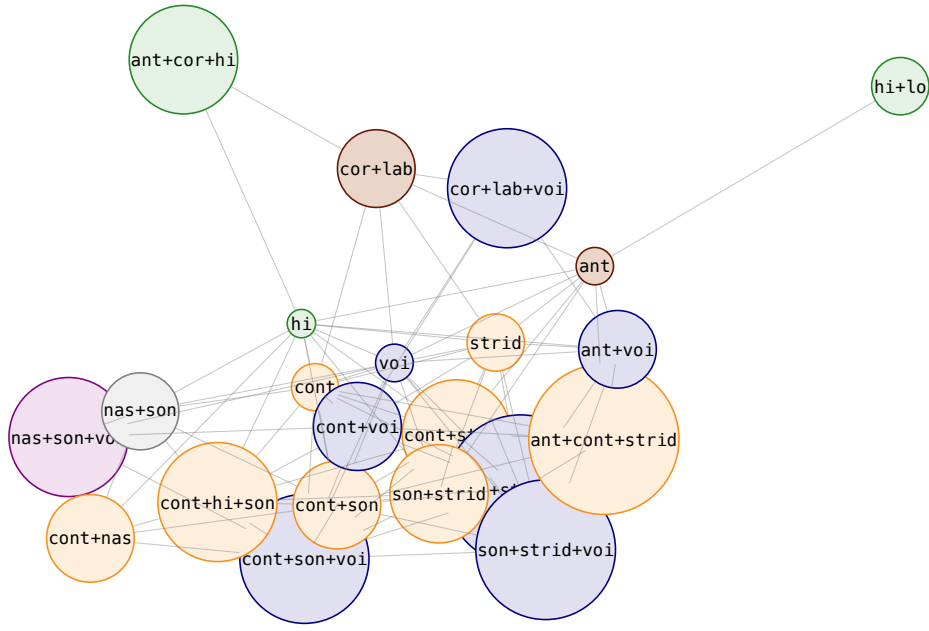


Figure 3: Operator graph G_O of Indo-European. Nodes are operators (the 22 with highest support), sized by frequency and coloured by change type (voicing, fricativization, palatalization/height, place, prenasalization). An edge joins u, v when their composition $u \oplus v$ is itself an operator of the repertoire — so this is the graph of *additive closure*. Dense connectivity is the visual face of the composition index $C(O)$.

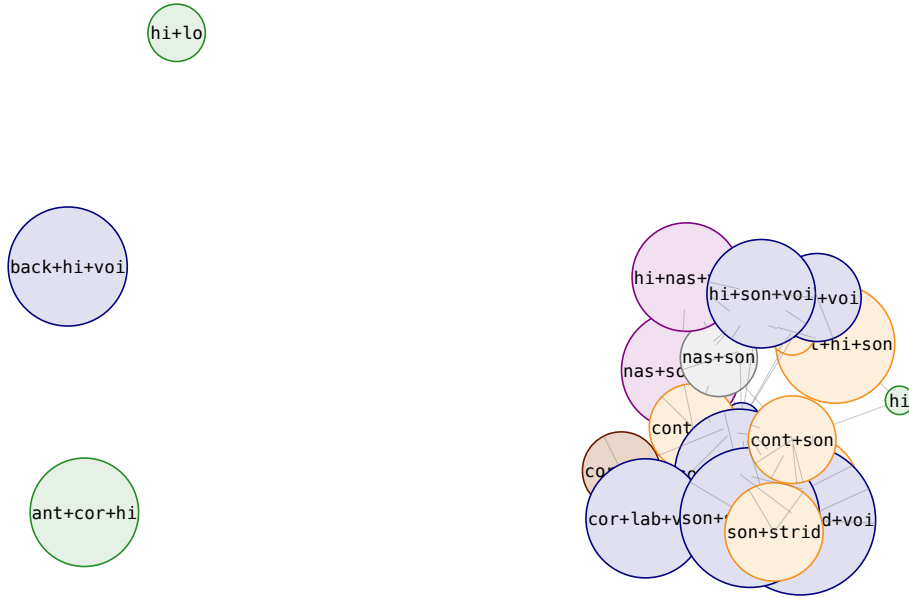


Figure 4: Operator graph G_O of Austronesian. Nodes are operators (the 22 with highest support), sized by frequency and coloured by change type (voicing, fricativization, palatalization/height, place, prenasalization). An edge joins u, v when their composition $u \oplus v$ is itself an operator of the repertoire — so this is the graph of *additive closure*. Dense connectivity is the visual face of the composition index $C(O)$.

Then the language graph G_L , built *only* from operator distributions (Jensen–Shannon distance), with nodes coloured — after the fact — by genealogical branch. Whether the colours cluster is a direct, visual version of the question in Chapters 14–15: how much genealogy do the distributions alone carry?

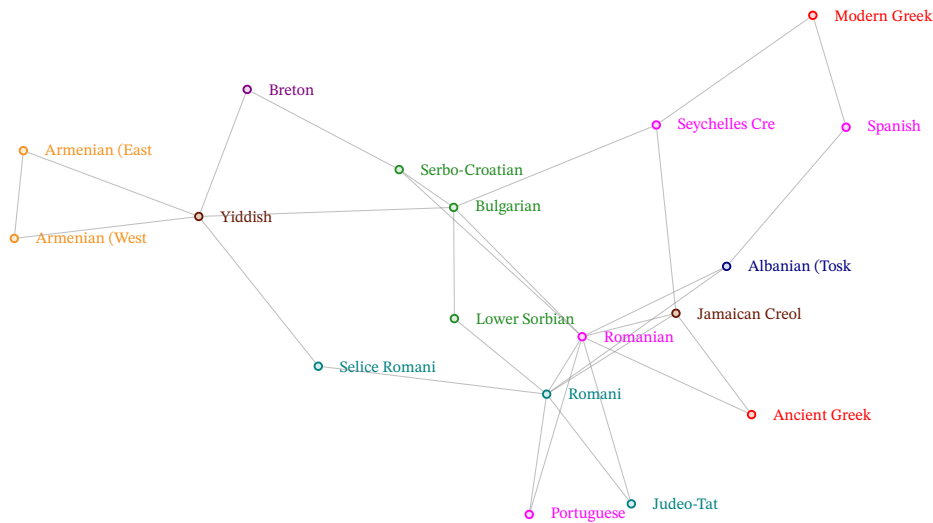


Figure 5: Language graph G_L of Indo-European. Each node is a language, coloured by its Glottolog branch; two languages are joined when one is among the other’s two nearest neighbours in *operator-distribution* distance (Jensen–Shannon). Colour was *not* used to build the graph — only the operator frequencies were. Whether same-colour nodes cluster is a visual test of how much genealogy the distributions alone carry (cf. Chapters 14, 18). Branches: **Albanian**, **Armenian**, **Balto-Slavic**, **Celtic**, **Germanic**, **Graeco-Phrygian**, **Indo-Iranian**, **Italic**.

22. Implications for AI language models (*speculative*)

This chapter is openly speculative. Two threads. First, representation: if human phonological systems occupy sparse, additively-structured, few-effective-dimensional regions of a feature space, that is a statement about the *inductive biases* a good model of sound structure might want — not one-hot segment inventories but feature-additive codes with a strong low-rank/affine prior. Second, methodology: the project’s core stance — *discover structure first, bring in the ground-truth labels only to test* — is exactly the discipline that guards machine-learning evaluation against leakage. A model that “recovers Grimm’s law” is impressive only if the law was never in its inputs; the historical-holdout protocol (Chapter 20) is a template for such honest evaluation of language models on historical and typological tasks. We make no stronger claim than that these are worth trying.

23. A program, not a study

This is one study; the object it opens is large. Persistent topology (build a complex with a triangle whenever $u \oplus v = w$ and watch cycles and voids across a frequency threshold), spectral analysis of G_S/G_O , and multilayer graphs (a layer per branch, period, context) are all natural next instruments — none of which reduce the geometry to a single number. And two things stay reserved on principle, to be brought in only as external partitions once the structure is mapped: protoforms, and OAS. The question for OAS is precise and non-circular: *do the dimensions, communities, or latent factors discovered from IPA and documented correspondences spontaneously recover groupings compatible with OAS?*

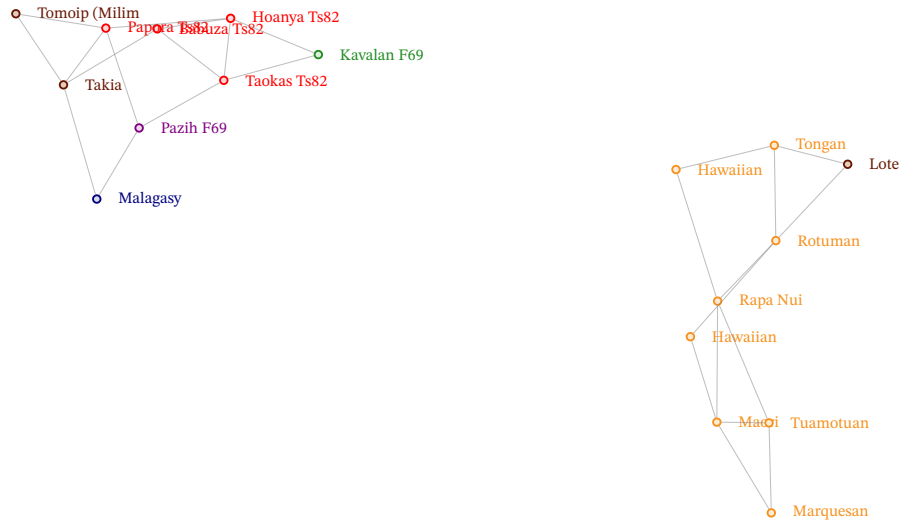


Figure 6: Language graph G_L of Austronesian. Each node is a language, coloured by its Glottolog branch; two languages are joined when one is among the other's two nearest neighbours in *operator-distribution* distance (Jensen–Shannon). Colour was *not* used to build the graph — only the operator frequencies were. Whether same-colour nodes cluster is a visual test of how much genealogy the distributions alone carry (cf. Chapters 14, 18). Branches: **Basap-Greater Barito**, **Central Pacific linkage**, **East Formosan**, **Northwest Formosan**, **Western Oceanic linkage**, **Western Plains Austronesian**.

If they do, that is independent evidence; if they do not, it lets the classes be revised without circularity. Either way, OAS earns its place by *surviving a blind test*, never by being assumed.

Appendices

Appendix A — Glossary of symbols

symbol	meaning
$\phi(a)$	panphon feature vector of segment a
$\Delta(a, b) = \phi(a) \oplus \phi(b)$	operator = symmetric feature-difference (the signature)
O (or O_L)	repertoire: the set of observed operator types
$\langle O \rangle$	subspace generated by O over $\mathbb{F}_2$; $r = \dim\langle O \rangle$ (rank)
$\langle O \rangle \setminus O$	the holes: generable but unattested operators
U_S	realizable universe: $\{\Delta(a, b) : a, b \in S\}$, $S =$ segment inventory
Ω_D	opportunity universe: differences occurring in aligned corpus positions
$\rho_{\text{alg}}, \rho_{\text{seg}}, \rho_{\text{opp}}$	occupancy vs span / realizable / opportunity
$C(O), C_{\Omega}(O)$	composition-realization index (raw / conditioned on opportunity)
$\tau(O), \kappa(O), E(O)$	triple density, doubling constant, additive energy
$K_{\cap}, K_{1/2}, K_{\alpha}$	branch nuclei (all / majority / threshold); $H_k, p(o)$
$P_L(o), H, N_{\text{eff}}$	persistence operator distribution, entropy, effective number of operators
$T = (a, b, c, p, \ell_1, \ell_2, w)$	the future <i>directed</i> transformation; $\sigma(T) = \Delta(a, b)$

Appendix B — Catalog of operations (**make** targets)

target	chapter	what it computes
make family FAMILY="X"	5	correspondences (LexStat + alignment) $\rightarrow$ repertoire
make algebra FAMILY="X"	8, 13	atoms, rank/corank, XOR-break, $C(O)$
make repr-control FAMILY="X"	8	representation control (O vs U_S rank/dependencies)
make universes FAMILY="X"	9	three universes, three occupancies, C_{Ω}
make nulls FAMILY="X"	10	six nested nulls for $C(O)$ with p_{MC} + bootstrap
make additive FAMILY="X"	10, 11, 12	τ, κ, E ; matroid circuits; hole geometry
make distributions FAMILY="X"	14	$P_L(o)$, entropy, N_{eff} , mutual information
make superposition FAMILY="X"	15	branch decomposition, grouping null (+ TF_RAREFY, TF_NULLBRANCH)
make cognate-eval	5	LexStat vs expert cognacy (pair- and type-level)
make regimes	17	four corpus regimes $D_G/D_L/D_C/D_R$ + loans
make chains FAMILY="X"	—	preferred change corridors (monotone dimensions)

Appendix C — Reproducibility

All results run from `src/` against `data/db/transf.db` (built by `make family`). Datasets: Lexibank (IE 304 / AN 978 languages available; per-family top languages by form count), IE-CoR (iecor, ex-

pert cognacy + segments), Glottolog classification (cached in `data/glottolog_classification.csv`, 26 748 paths). Feature matrix: panphon (ternary values; primary feature set of 12 including lab). Cognacy: LexStat, threshold 0.55, scorer 100 runs. Alignment: Needleman–Wunsch with panphon feature-fraction cost. Thresholds: operator support ≥ 30 (family), ≥ 8 (branch); operators of 1–3 features. Exact per-family counts are printed by each script. Seeds are fixed in the null/bootstrap code for reproducibility.

Appendix D — Revision history

This manual descends from an internal report that went through three external review rounds. Round 1 corrected the conflation of representation with data (the “group under XOR” is a property of the encoding, not of languages). Round 2 added the nucleus/superposition analysis and the cognate-vs-conceptual validation. Round 3 demanded, and received, the realizable/opportunity universes, the nested nulls, the basis-dependence distinction, the two-level cognacy metrics, and the grouping null (which overturned the round-2 “IE more differentiated” reading). Retractions along the way — most notably “nasality is not an independent movement” (failed a null) and “branch richness differs” (a sampling artifact) — are kept here rather than in the body, so the argument can be read in its corrected form.

Appendix E — Data supplement

Full operator tables, branch-by-branch repertoires, the exhaustive segmental graphs G_S for both families, and the per-chapter raw outputs live under `docs/` and `data/db/`, regenerated by the targets in Appendix B. Full-page figures and the interactive graphs are collected here as the manual matures.